Words or Vision: Do Vision-Language Models Have Blind Faith in Text?

Anonymous CVPR submission

Abstract

Vision-language models are increasingly deployed in settings where an image is accompanied by auxiliary text, such as retrieved passages, webpage content, OCR, or agent state. In practice, this extra text can be helpful, noisy, or misleading, yet most existing vision-language benchmarks do not directly test how models arbitrate when text conflicts with visual evidence. We introduce BFiT, a benchmark for measuring text over-reliance in VLMs through controlled perturbations of auxiliary text. BFiT augments 5,500 base examples from VQAv2, DocVQA, MathVista, and a brand-detection task derived from KnowPhish with three text conditions, Match*,* Corruption*, and* Irrelevance*, yielding 16,500 multimodal test cases with fixed images and questions. We evaluate ten representative proprietary and open models using deterministic decoding, and report condition accuracy, Macro Accuracy, Normalized Corruption Accuracy, and a Text Preference Ratio (*TPR*) that quantifies whether multimodal predictions track text rather than image under conflict. Results reveal substantial blind faith in text for many open models: on VQAv2 corruption, Claude Sonnet attains 68.17% accuracy with 9.58%* TPR*, while LLaVA-NeXT-7B falls to 28.69% with 85.52%* TPR*. On brand detection, GPT-4o remains stable under corrupted text (89.76%, 0.83%* TPR*), whereas Molmo-7B-D drops to 41.44% (60.40%* TPR*). Cautionary instructions help only modestly, larger open models remain vulnerable, and supervised fine-tuning with conflict-aware data greatly improves corruption robustness, albeit with some clean-performance trade-offs. BFiT is designed as a targeted evaluation protocol for measuring multimodal arbitration rather than overall vision-language ability.*

1. Introduction

Modern multimodal systems rarely operate on images alone. Retrieval-augmented generation pipelines attach retrieved passages, document understanding systems combine screenshots with OCR or layout text, and browser or desktop agents receive structured interface text together with pixels. In all of these settings, the additional text can be useful, irrelevant, or actively misleading. A robust VLM should exploit helpful text while still grounding its final answer in the image when the two modalities disagree.

Existing benchmarks only partially capture this problem. Broad capability benchmarks such as MMBench, SEED-Bench, and MMMU measure general multimodal performance, but they do not systematically intervene on auxiliary text while holding the underlying visual question fixed [9, 13, 19]. Vision-indispensable benchmarks such as MM-Star and benchmark-auditing efforts such as SugarCrepe and "Making the V in VQA Matter" highlight shortcut learning and missing visual grounding, but they are not designed to study how VLMs respond to supportive versus conflicting external text [5, 7, 8]. This gap matters because the failure mode is operationally different: a model may be visually capable in isolation yet still answer incorrectly once exposed to plausible but wrong text.

We address this gap with BFiT, a benchmark for *modality arbitration under auxiliary-text perturbations*. Starting from four vision-centric tasks, we keep the image and question fixed and inject one of three short text conditions: *Match* text that supports the correct answer, *Corruption* text that is plausible but steers toward a different answer, and *Irrelevance* text sampled from an unrelated corpus. The resulting setup isolates a single evaluation question: when text and image send different signals, which modality does the model trust?

BFiT yields three main empirical findings. First, many open models display strong text preference under corruption, with severe drops in normalized accuracy and high TPR. Second, this vulnerability is highly task dependent: it is especially consequential for real-world brand detection, where corrupted auxiliary text can erase most of the benefit of otherwise strong base performance. Third, prompting alone is weak mitigation, whereas conflict-aware supervised fine-tuning substantially improves robustness and partially transfers to held-out tasks.

Contributions. We make four contributions. (1) We introduce BFiT, a 16.5K-example benchmark built from 5.5K base samples across VQAv2, DocVQA, MathVista, and brand detection, specifically designed to evaluate multimodal arbitration under supportive, misleading, and irrelevant text. (2) We define a compact evaluation protocol centered on condition accuracy, Macro Accuracy, Normalized Corruption Accuracy, and TPR. (3) We provide a com-

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parative study over ten VLMs and show large robustness gaps between proprietary and open models. (4) We analyze several factors that influence text bias, including prompting, language-model scale, text relevance, token order, and conflict-aware fine-tuning.

2. Related Work

Benchmarking and auditing VLMs. The recent VLM literature has emphasized broad, task-diverse evaluation through benchmarks such as SEED-Bench, MMBench, MMMU, and MathVista [9, 13, 14, 19]. These benchmarks are valuable for measuring capability coverage, but they generally score a single prompt per example and therefore provide limited leverage for studying whether a model overtrusts auxiliary text. Other work has shown that many multimodal benchmarks are not fully vision dependent, motivating more carefully curated evaluations such as MMStar [5]. SugarCrepe similarly demonstrates that even compositionality benchmarks can be hackable without careful negative construction [8]. BFiT complements these efforts by keeping the base task intact and intervening only on the external text presented at inference time.

Language priors, conflicting evidence, and noisy context. Shortcut exploitation has long been a central concern in visual question answering; VQAv2 itself was introduced to make image understanding matter more strongly [7]. In language-only settings, noisy and conflicting context has also been studied in RAG and knowledge-conflict benchmarks [4, 18]. Recent work on cross-modality knowledge conflict in VLMs focuses on disagreements between parametric knowledge sources inside the model and proposes mitigation strategies [20]. Our setup is complementary: we study *explicit auxiliary text* supplied at inference time, which is common in deployed systems and directly controllable by benchmark design. This distinction matters because a model can be vulnerable even when its underlying visual capability is strong.

Representative models. We evaluate both proprietary and open systems, including GPT-family models [2], Claude-family models [3], LLaVA-NeXT [11], Qwen2-VL [17], Molmo [6], and Phi-3.5-Vision [1]. BFiT is not meant to rank these models on overall multimodal ability; rather, it isolates a robustness axis that standard leaderboards do not separately expose.

3. Benchmark Design

3.1. Base Tasks

BFiT starts from four vision-centric tasks chosen to cover natural images, documents, mathematical diagrams, and security-relevant web screenshots:

- **VQAv2** [7]: 1,000 validation examples.
- **DocVQA** [15]: 1,000 validation examples emphasizing charts and tables.
- **MathVista** [14]: 1,000 examples from the minitest split.
- **Brand detection** derived from **KnowPhish** [10]: 2,500 webpage screenshots used for brand recognition.

This yields 5,500 base image-question pairs. Each pair is evaluated once in its original form and three more times with injected text, producing 16,500 augmented multimodal test examples.

3.2. Constructing Auxiliary Text

For each base example, BFiT constructs three short text variants.

- **Match**: text that is sufficient to support the gold answer without relying on the image.
- **Corruption**: text that is fluent and task-relevant but leads toward a different answer than the image supports.
- **Irrelevance**: unrelated text sampled from WikiText [16].

Matched and corrupted descriptions are generated with GPT-4o from the image-question-answer tuple. The key design principle is to change only the auxiliary text while preserving the image, question, and evaluation target. This makes the benchmark diagnostic: any behavior change across conditions must come from how the model responds to the added text rather than from a change in the underlying task instance.

3.3. Prompting and Packaging

Each benchmark input follows the same structure:

```
image + optional cautionary instruction + auxiliary
text + question
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The optional instruction warns that the text may be inaccurate or irrelevant and that the model should prioritize visual evidence when the modalities disagree. We study this instruction as a lightweight mitigation rather than making it part of the default benchmark. All reported evaluations use deterministic decoding with temperature 0.

3.4. Evaluated Models

We evaluate ten representative VLMs: GPT-4o mini, Claude Haiku, GPT-4o, Claude Sonnet, LLaVA-NeXT-7B, LLaVA-NeXT-13B, LLaVA-NeXT-34B, Phi-3.5-Vision, Molmo-7B-D, and Qwen2-VL-7B [1–3, 6, 11, 17]. The mix is intentionally heterogeneous: some models are instruction-tuned proprietary APIs, while others are open models whose training recipe and backbone scale vary substantially.

3.5. Metrics

We report four metrics.

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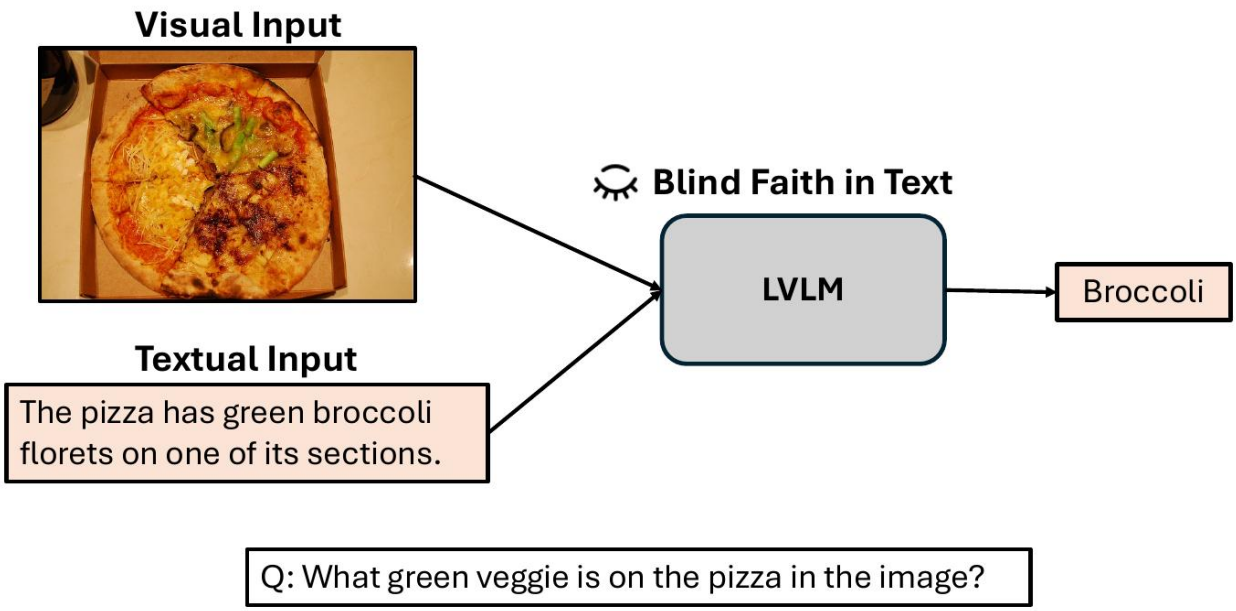

Figure 1. BFiT benchmark overview. For each base image-question pair, we append auxiliary text in one of three controlled conditions: *Match*, which supports the correct answer; *Corruption*, which is plausible but misleading; and *Irrelevance*, which is unrelated to the visual content. The benchmark measures whether the model's prediction remains image-grounded or shifts toward the textual context when the modalities agree, conflict, or decouple.

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Condition Accuracy. Accuracy under each condition, including the original base setting with no injected text.

Macro Accuracy. The mean of the three injected-text conditions:

$$\text{Macro} = \frac{1}{3}\left(A_{\text{match}} + A_{\text{corr}} + A_{\text{irr}}\right). \tag{1}$$

This avoids over-interpreting models that are strong only when text is supportive.

Normalized Corruption Accuracy. To isolate robustness relative to a model's clean-task strength, we normalize corruption accuracy by base accuracy:

$$\text{Norm} = 100 \cdot \frac{A_{\text{corr}}}{A_{\text{base}}}. \tag{2}$$

Values near 100 indicate that misleading text has little net effect; values above 100 indicate that corrupted text was occasionally beneficial.

Text Preference Ratio (TPR). For corruption examples, we additionally compare the multimodal answer to the model's text-only and image-only answers on the same question. TPR is the fraction of cases where the multimodal response is more consistent with the text-only answer than the image-only answer, with partial-overlap handling for non-exact matches. Lower is better: a high TPR means the model tends to follow misleading text instead of visual evidence.

4. Benchmark Validation

Sanity check: does the injected text create the intended regimes? Table 1 evaluates text-only inputs on VQAv2. Matched descriptions are highly answerable without images: GPT-4o reaches 90.07% and Claude Sonnet 84.39%. In contrast, corrupted text leads to much lower text-only accuracy, 17.59% for GPT-4o and 16.17% for Claude Sonnet. Irrelevant text is also substantially weaker, though not uniformly near zero, ranging from 18.67% to 35.40%. This is the intended behavior: matched text often contains enough signal to solve the task, while corrupted and irrelevant text do not simply reproduce the gold answer.

	VQAv2			
Model	Base	Match	Corruption	Irrelevance
Claude Sonnet	66.88	84.39	16.17	24.39
GPT-4o	78.39	90.07	17.59	18.67
Molmo-7B-D	76.33	88.98	18.74	35.40

Table 1. Text-only accuracy (%) across different models. It provides a sanity check for the constructed text when matched, corrupted, or irrelevant.

What BFiT measures. BFiT measures a specific failure mode: robustness to auxiliary text that is supportive, misleading, or unrelated. It does *not* attempt to measure overall multimodal competence, long-context retrieval quality, or open-world factuality. This narrower scope is deliberate. By holding the underlying visual question fixed and changing only the added text, BFiT provides a clean test of modality arbitration.

The metric suite is designed to make distinct failure modes visible. Base or matched-text accuracy alone can reward models that follow text whenever it is present, while corruption accuracy alone can over-credit models that simply ignore all auxiliary text, including useful evidence. Reporting Macro Accuracy, Normalized Corruption Accuracy, and TPR together distinguishes robust multimodal arbitration from indiscriminate text-following and from blanket text rejection.

This diagnostic separation is important for benchmark validity. A text-copying model can look strong in the Match condition yet fail badly under Corruption and exhibit high TPR, whereas a text-ignoring model can preserve corruption robustness but leave useful evidence unused and therefore underperform on Match and Macro Accuracy. Because these signatures diverge, BFiT reveals not only whether a system succeeds, but *how* it uses auxiliary text. That property is especially relevant for deployments that combine images with OCR, retrieved passages, or agent state, where the desired behavior is selective reliance rather than unconditional trust or unconditional dismissal.

5. Benchmark Results

Table 2 reports the main corruption results on VQAv2, DocVQA, and MathVista. The benchmark reveals a consistent divide between proprietary models and most open models.

VQAv2. Claude Sonnet is the most robust model under corrupted text, achieving 68.17% corruption accuracy with a TPR of only 9.58%, and even slightly exceeding its base accuracy after normalization (101.93). GPT-4o is next best at 70.75% corruption accuracy and 27.09% TPR. Open models are much more vulnerable: LLaVA-NeXT-7B drops from 79.45% base accuracy to 28.69% under corruption, with 36.10 normalized accuracy and 85.52% TPR. Qwen2-VL-7B is the strongest open model on base accuracy (85.51%), but still falls to 50.79% under corruption with 59.41 normalized accuracy.

DocVQA. The robustness gap is even sharper on documents. Claude Sonnet reaches 84.60% corruption accuracy and 3.21% TPR, while GPT-4o attains 73.60% and 17.96%. LLaVA-NeXT variants collapse to 10.00%–15.10% corruption accuracy with TPR above 82%. Qwen2-VL-7B remains the strongest open model here as well, but its corruption accuracy still drops from 90.50% to 57.50%.

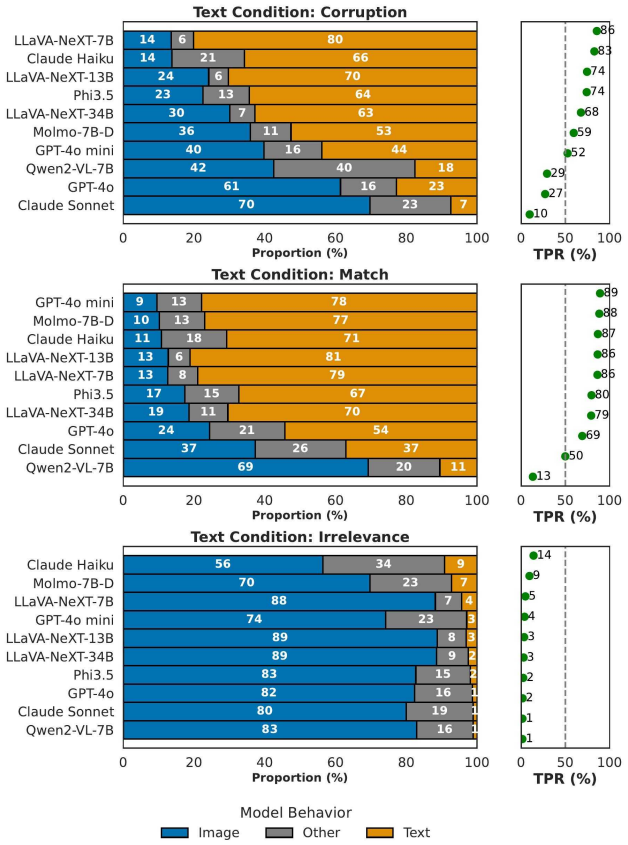

Figure 2. Prediction behavior across text conditions. Corrupted text causes a pronounced shift toward text-consistent predictions for many open models, while proprietary models remain substantially more image-consistent.

MathVista. MathVista is difficult for all models, yet the same pattern remains. Claude Sonnet again leads with 49.30% corruption accuracy, 87.57 normalized accuracy, and 29.14% TPR. Most open models are below 33% corruption accuracy and have TPR above 60%, indicating that misleading textual cues strongly affect their answers on visually grounded math problems.

Brand detection is a particularly consequential stress test. Table 3 shows that corrupted text can be dangerous even for tasks with high base accuracy. GPT-4o remains essentially unchanged under corruption, rising slightly from 88.68% to 89.76% with a TPR of 0.83%. Claude Sonnet is similarly stable at 90.24% corruption accuracy and 0.96% TPR. In contrast, Molmo-7B-D falls from 87.44% to 41.44% and reaches only 47.39 normal-

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Model	VQAv2				DocVQA				MathVista			
	Base ↑	Corruption ↑	Norm ↑	TPR ↓	Base ↑	Corruption ↑	Norm ↑	TPR ↓	Base ↑	Corruption ↑	Norm ↑	TPR ↓
GPT-4o mini	69.82	51.55	73.83	52.42	69.40	38.20	55.04	52.07	52.30	23.90	45.70	80.28
Claude Haiku	50.08	25.54	50.99	82.70	68.80	40.20	58.43	47.67	41.00	19.80	48.29	77.42
GPT-4o	78.39	**70.75**	90.25	27.09	85.00	73.60	86.59	17.96	**58.90**	41.20	69.95	48.98
Claude Sonnet	66.88	68.17	**101.93**	**9.58**	87.00	**84.60**	**97.24**	**3.21**	56.30	**49.30**	**87.57**	**29.14**
LLaVA-NeXT-7B	79.45	28.69	36.10	85.52	53.60	10.00	18.60	87.77	35.80	19.70	54.97	84.19
LLaVA-NeXT-13B	81.02	37.61	46.40	74.43	57.70	11.00	19.10	86.84	36.20	20.60	56.89	80.83
LLaVA-NeXT-34B	82.96	42.87	51.70	67.56	64.00	15.10	23.61	82.69	34.00	21.70	61.98	67.64
Phi3.5	75.65	35.23	46.50	74.05	78.20	50.50	64.60	40.51	43.10	22.20	51.47	80.20
Molmo-7B-D	76.33	49.29	64.50	59.40	74.00	38.40	51.90	57.20	44.90	32.90	73.27	60.63
Qwen2-VL-7B	**85.51**	50.79	59.41	29.22	**90.50**	57.50	63.63	37.41	55.40	28.90	52.18	70.23

Table 2. Performance (%) under corrupted text, reported as Base Accuracy, Corruption Accuracy, Normalized Corruption Accuracy (Norm), and Text Preference Ratio (TPR). **Bold** marks the best result and underlining marks the second best.

ized accuracy. LLaVA-NeXT-7B drops to 55.32%, and Phi-3.5 to 60.68%. This result suggests that auxiliary-text conflicts are not merely academic; they can materially affect security-relevant settings where web text may be adversarial or misleading.

Model	Brand Detection			
	Base ↑	Corruption ↑	Norm ↑	TPR ↓
GPT-4o mini	88.84	84.8	95.44	7.48
Claude Haiku	84.40	78.72	93.27	6.44
GPT-4o	88.68	**89.76**	101.22	**0.83**
Claude Sonnet	**90.20**	90.24	100.04	0.96
LLaVA-NeXT-7B	78.60	55.32	70.39	59.17
LLaVA-NeXT-13B	83.00	60.00	72.29	40.65
LLaVA-NeXT-34B	66.28	53.52	80.77	23.49
Phi3.5	84.40	60.68	71.90	50.45
Molmo-7B-D	87.44	41.44	47.39	60.40
Qwen2-VL-7B	89.68	86.48	96.43	2.99

Table 3. Performance on brand detection, reported as Base Accuracy, Corruption Accuracy, Normalized Corruption Accuracy (Norm), and Text Preference Ratio (TPR). **Bold** marks the best result and underlining marks the second best.

6. Analysis

6.1. Prompting Helps Little; Scale Helps Somewhat

Figure 3 summarizes several factors affecting text bias. The first is prompting. A cautionary instruction improves corruption accuracy only modestly on VQAv2: LLaVA-NeXT-7B rises from 28.69% to 34.27%, and Qwen2-VL-7B from 50.79% to 54.78%. This is a real effect, but it is far smaller than the underlying robustness gap.

The second is model scale. Within LLaVA-NeXT, increasing the language model from 7B to 34B reduces TPR, but does not eliminate text bias. On VQAv2, TPR decreases from 85.52% to 67.56%; on brand detection it decreases from 59.17% to 23.49%. Scaling therefore helps, but it does not make the model reliably image-first under conflict.

6.2. Text Relevance and Token Order Matter

The third factor is *text relevance*. Figure 3(c) shows that auxiliary text that appears semantically related to the task induces more text preference than obviously irrelevant content, even when the related text is not helpful for solving the instance. This suggests that the failure mode is not merely distraction by extra tokens; it is selective trust in task-looking language.

The fourth factor is *token order*. Figure 3(d) shows that, for Phi-3.5, placing text before image tokens increases text bias. This effect is consistent with the broader observation that multimodal generation can depend strongly on the ordering and formatting of inputs, especially when the model is uncertain [12].

6.3. Conflict-Aware Fine-Tuning Is More Effective Than Prompting

Table 4 compares the original models, the instruction-augmented variants, and supervised fine-tuning (SFT) with conflict-aware text augmentation on VQAv2. The gains are substantial. For LLaVA-NeXT-7B, corruption accuracy improves from 28.69% to 71.25%, and Macro Accuracy from 66.81% to 78.71%. For Qwen2-VL-7B, corruption accuracy rises from 50.79% to 82.72%, and Macro Accuracy from 75.75% to 84.58%.

These gains are not free. Both models lose some clean and matched-text performance after SFT. LLaVA-NeXT-7B drops from 79.45% to 77.48% on the base setting and from 92.32% to 87.56% on matched text. Qwen2-VL-7B similarly declines from 85.51% to 84.18% on the base setting and from 92.76% to 87.01% on matched text. The mitigation is therefore meaningful but not costless, and Macro Accuracy is a more faithful summary than looking at corruption alone.

The fine-tuning effect also generalizes beyond the in-distribution VQAv2 setup. On held-out tasks, LLaVA-NeXT-7B's Macro Accuracy improves from 51.07% to 56.17% on DocVQA and from 46.44% to 72.29% on brand detection, while Qwen2-VL-7B improves from 80.83% to

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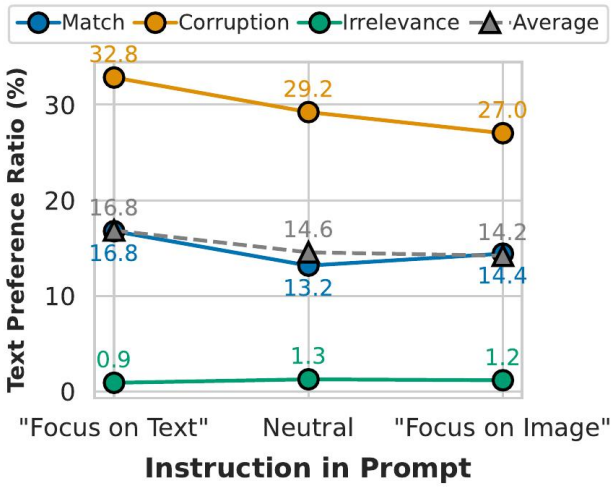
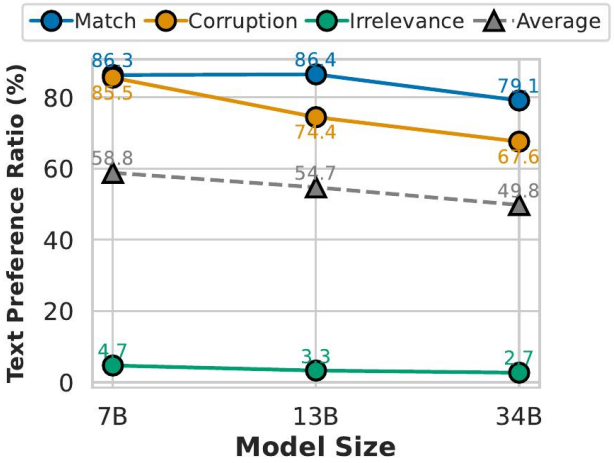
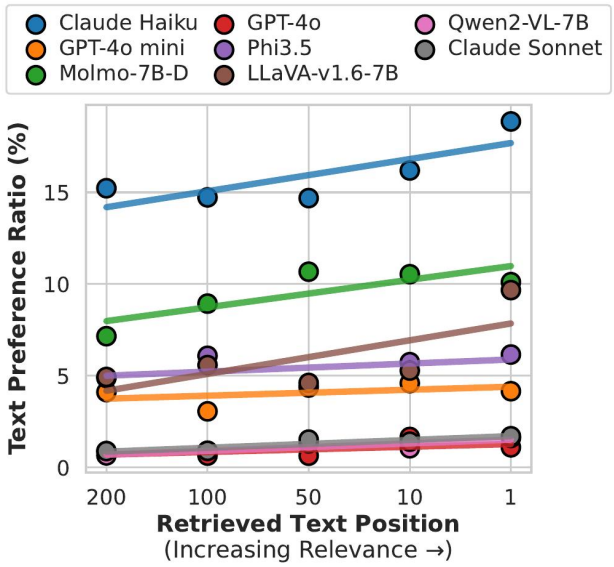


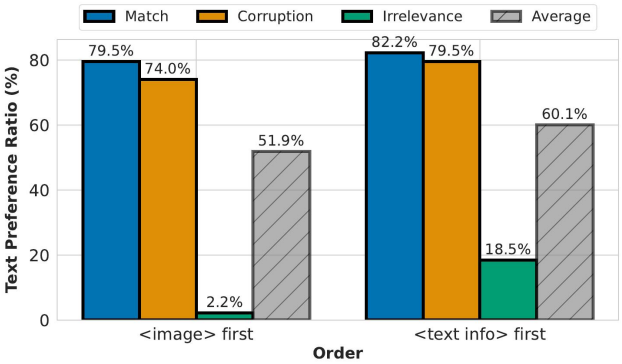
(a) Cautionary prompting.



(b) Language-model scale.



(c) Text relevance.



(d) Token order.

Figure 3. Factors affecting text bias. Prompting has only limited effect; larger open models are somewhat less text-biased but still vulnerable; seemingly relevant auxiliary text increases text preference; and, for Phi-3.5, placing text tokens before image tokens increases text bias.

88.97% on DocVQA, from 53.87% to 57.17% on MathVista, and from 81.85% to 88.75% on brand detection. These results indicate that conflict-aware training is not merely overfitting to one dataset's corruption pattern.

	VQAv2				
Model	Base ↑	Match ↑	Corruption ↑	Irrelevance ↑	Macro ↑
LLaVA-NeXT-7B	**79.45**	**92.32**	28.69	**79.43**	66.81
Instruction	**79.45**	92.25	34.27	78.15	68.22
SFT	77.48	87.56	**71.25**	77.32	**78.71**
Qwen2-VL-7B	**85.51**	**92.76**	50.79	83.70	75.75
Instruction	**85.51**	92.62	54.78	82.82	76.74
SFT	84.18	87.01	**82.72**	**84.00**	**84.58**

Table 4. In-distribution VQAv2 comparison between the original models, caution-instruction variants, and conflict-aware supervised fine-tuning (SFT).

7. Discussion

BFiT exposes a failure mode that conventional multimodal leaderboards largely hide. A model can look strong on base accuracy yet become brittle once it receives plausible extra text. This matters in exactly the settings where VLMs are now being deployed: agent systems, retrieval-augmented multimodal assistants, document processing, and webpage understanding.

Protocol implications.

- Rankings depend strongly on condition. On VQAv2, Qwen2-VL-7B is competitive on base accuracy but drops under Corruption because it still tracks misleading text, whereas Claude Sonnet is more stable when text conflicts with the image. Single-number leaderboards therefore miss an important robustness axis, and TPR should be reported together with all three text conditions.
- Because BFiT fixes the image and question across conditions, regressions can be attributed to modality arbitration rather than changes in task mix. This makes the benchmark useful for ablations, safety checks, and release gating in text-rich deployments.
- The benchmark also separates selective grounding from blanket text rejection: ignoring all auxiliary text may protect corruption accuracy while sacrificing Match and Macro Accuracy.

Scope boundaries.

- BFiT evaluates short injected text rather than full retrieval pipelines or multi-turn interaction.
- The matched and corrupted texts are generated, so stylistic artifacts may remain even though text-only sanity checks suggest that the intended regimes are broadly realized.
- The benchmark covers four tasks rather than the entire multimodal landscape.

We therefore view BFiT as a targeted stress test for modality arbitration, not a complete robustness benchmark for all multimodal systems.

The mitigation results also suggest a useful hypothesis about the source of blind faith in text. Prompting helps only slightly, whereas conflict-aware supervised fine-tuning helps substantially. This pattern is consistent with a training-data imbalance story: VLMs are exposed to vast amounts of text and image-text data, but comparatively little supervision that explicitly teaches them to distrust plausible textual context when it conflicts with visual evidence. We present this as an interpretation supported by the empirical pattern, not as a settled causal claim.

8. Conclusion

We presented BFiT, a benchmark for evaluating whether VLMs over-trust auxiliary text relative to visual evidence. By augmenting 5.5K base examples across four tasks with matched, corrupted, and irrelevant text, BFiT provides a controlled and reproducible way to measure multimodal arbitration. Across ten representative models, we find that misleading text causes major failures for many open models, while stronger proprietary models are markedly more robust. Prompt-based warnings offer limited mitigation, but conflict-aware fine-tuning materially improves robustness and partially transfers across tasks. We hope BFiT helps shift multimodal evaluation toward not only what models

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can solve, but also what evidence they trust when modalities disagree.

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References

[1] Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Jianmin Bao, Harkirat Behl, et al. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024.

[2] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

[3] Anthropic. Claude 3 model card, 2024. https://www-cdn.anthropic.com/de8ba9b01c9ab7cbabf5c33b80b7bbc618857627/Model_Card_Claude_3.pdf.

[4] Jiawei Chen, Hongyu Lin, Xianpei Han, and Le Sun. Benchmarking large language models in retrieval-augmented generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 17754–17762, 2024.

[5] Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, et al. Are we on the right way for evaluating large vision-language models? *arXiv preprint arXiv:2403.20330*, 2024.

[6] Matt Deitke, Christopher Clark, Sangho Lee, Rohun Tripathi, Yue Yang, Jae Sung Park, Mohammadreza Salehi, Niklas Muennighoff, Kyle Lo, Luca Soldaini, et al. Molmo and pixmo: Open weights and open data for state-of-the-art multimodal models. *arXiv preprint arXiv:2409.17146*, 2024.

[7] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6904–6913, 2017.

[8] Cheng-Yu Hsieh, Jieyu Zhang, Zixian Ma, Aniruddha Kembhavi, and Ranjay Krishna. Sugarcrepe: Fixing hackable benchmarks for vision-language compositionality. In *Thirty-Seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023.

[9] Bohao Li, Rui Wang, Guangzhi Wang, Yuying Ge, Yixiao Ge, and Ying Shan. Seed-bench: Benchmarking multimodal llms with generative comprehension. *arXiv preprint arXiv:2307.16125*, 2023.

[10] Yuexin Li, Chengyu Huang, Shumin Deng, Mei Lin Lock, Tri Cao, Nay Oo, Bryan Hooi, and Hoon Wei Lim. Knowphish: Large language models meet multimodal knowledge graphs for enhancing reference-based phishing detection. *arXiv preprint arXiv:2403.02253*, 2024.

[11] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge, 2024.

[12] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

[13] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. Mmbench: Is your multi-modal model an all-around player? In *European Conference on Computer Vision*, pages 216–233. Springer, 2025.

[14] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*, 2023.

[15] Minesh Mathew, Dimosthenis Karatzas, and CV Jawahar. Docvqa: A dataset for vqa on document images. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 2200–2209, 2021.

[16] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models, 2016.

[17] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.

[18] Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. *arXiv preprint arXiv:2305.13300*, 2023.

[19] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. *arXiv preprint arXiv:2311.16502*, 2023.

[20] Tinghui Zhu, Qin Liu, Fei Wang, Zhengzhong Tu, and Muhao Chen. Unraveling cross-modality knowledge conflict in large vision-language models. *arXiv preprint arXiv:2410.03659*, 2024.

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